MARK PARAL

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PERSONAL STATEMENT:
Aeronautical and Astronautical Engineer seeking GNC opportunities in the aerospace sector!

EDUCATION:

Stanford University – Stanford, CA — Graduated: April 2025
Master of Science in Aeronautics and Astronautics — GPA: 3.94/4.00

Purdue University – West Lafayette, IN — Graduated with Distinction: May 2023
Bachelor of Science in Aeronautical and Astronautical Engineering — GPA: 3.90/4.00
Certificate in Entrepreneurship and Innovation

Relevant Coursework:
- Learning Based and Optimal Control – Convex Optimization – Decision Making Under Uncertainty – Spacecraft Attitude Dynamics – Trajectory Optimization – Robot Autonomy – Kalman Filtering

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EXPERIENCE:

Space Systems Engineer II – Elveo Mobile (Formerly Lynk Global) – Chantilly, VA — July 2025 – Present
- Developing satellite systems engineering simulation stack for analysis and optimization of constellation level power, thermal, and attitude control systems
- Created automated pipelines for definitive and predictive ephemeris production and distribution for spacecraft fleet
- Responsible for conjunction assessment and risk mitigation for a fleet of LEO satellites
- Developing and maintaining hardware-in-the-loop simulation infrastructure for in-orbit and in-development spacecraft
- Designing algorithms for spacecraft control and state estimation
- Recruiting and hiring for engineering roles

Graduate Researcher – Stanford Intelligent Systems Laboratory – Stanford, CA — Sept 2024 – Mar 2025
- Developed a custom 3D quadrotor simulation environment to experimentally evaluate a variety of control frameworks
- Created a Signed Distance Function (SDF) algorithm to characterize operating environments
- Implemented Augmented Lagrangian Iterative Linear Quadratic Regulator (AL-iLQR) and Sequential Convex Programming (SCP) trajectory generation and tracking frameworks
- Received CoDIT 2025 Best Paper award for "Improving the Resilience of Quadrotors in Underground Environments by Combining Learning-based and Safety Controllers"

Flight Dynamics Intern – Lynk Global – Falls Church, VA — June 2024 – Sept 2024
- Led development of a new state estimation architecture, applicable to both on-ground and in-orbit operations
- Implemented a high-fidelity propagator to improve state estimation accuracy and prediction of future trajectories
- Engineered post processing algorithms for precise state estimation and orbit determination

Aerospace Engineering Intern – Lynk Global – Falls Church, VA — May 2023 – Aug 2023
- Assembled hardware-in-the-loop (HWIL) setup for novel direct-to-phone satellites
- Developed software for handling CAN network interfaces between sensors, actuators, and dynamics simulator in C
- Built sensor simulators using microcontrollers
- Automated reaction wheel qualification testing

Undergraduate Research Assistant – Dr. Steven Son's Group – Zucrow Laboratories, IN — Aug 2022 – May 2023
- Operated test setups experimentally measuring the flexoelectric coefficient of a variety of materials
- Fabricated 3D-printed energetic test samples
- Reviewed published literature on flexoelectricity and its applications to energetic materials
- Performed t-test analytics on resulting data to determine the significance of results

Avionics Hardware Intern – Blue Origin – Seattle, WA May 2022 – Aug 2022
- Developed novel solution to enhance and expand sensor infrastructure on the New Shepard Propulsion Module
- Designed custom PCBs for signal conditioning and power distribution on the New Shepard launch vehicle
- Performed audit of New Shepard crew capsule sensors
- Wrote engineering work orders to resolve non-conformance reports

Avionics Intern – Firefly Aerospace – Austin, TX Jan 2022 – May 2022
- Developed custom hardware and software solution to automate qualification process for harnessing on all vehicles
- Constructed payload wiring diagrams for the Alpha launch vehicle in Siemens Capital software
- Created and led a test campaign to qualify securing devices used on multiple spacecraft
- Built custom Python executables to allow for quick updates to components in Siemens Capital software

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PROJECTS:

mako-sgp4 – *A theory-based SGP4 satellite propagator written in Rust* Oct 2025 – Present
- Developed my own open-source software package to propagate any satellite TLE or OMM to a desired position and velocity in time using the SGP4 algorithm implemented in Rust (https://github.com/markparal/mako-sgp4)
- Novel implementation of theory found in "History of Analytical Orbit Modeling in the U.S. Space Surveillance System" by Hoots et al with practical adjustments from "Revisiting Spacetrack Report #3" by Vallado et al
- Demo found at https://markparal.com/projects/mako-sgp4/demo/

TECHNICAL SKILLS:
- *Programming*: Rust – Python – C/C++ – MATLAB/Simulink – Julia
- *CAD*: SolidWorks – CATIA – NX – Fusion
- *PCB Design*: Altium – EAGLE – OrCAD – Siemens Capital
- *Other*: Level 2 High Powered Rocketry – Linux – Amateur Radio Technician – Soldering – STK/ODTK

ACHIEVEMENTS:
- International Conference on Control, Decision and Information Technologies (CoDIT) 2025 Best Paper Award
- 2023 Matthew Isakowitz Fellow
- Recipient of a Purdue AAE Undergraduate General Scholarship 2022 – 2023
- Recipient of the Joe Melroy Umbreit Scholarship 2020 – 2021
- Winner of the 2021 Purdue AAE 251 Northrop Grumman Rocket Design Contest